ER Diagram

- Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.

Step 1 : Identify the entity sets

- From the given question the entity sets identified are
  1. PERSON
  2. CAR
  3. ACCIDENT

Step 2 : Identify the relevant attributes

- The relevant attributes of PERSON entity set are
  - Person_Id
  - Person_Name
  - Address
- The relevant attributes of CAR entity set are
  - Engine_No
  - Model
  - Year
- The relevant attributes of ACCIDENT entity set are
  - Report_No
  - Location
  - Acc_Date

Step 3 : Identify the prime attributes

- The prime attribute of PERSON entity set is
  - Person_Id
- The prime attribute of CAR entity set is
  - Engine_No
- The prime attribute of ACCIDENT entity set is
  - Report_No

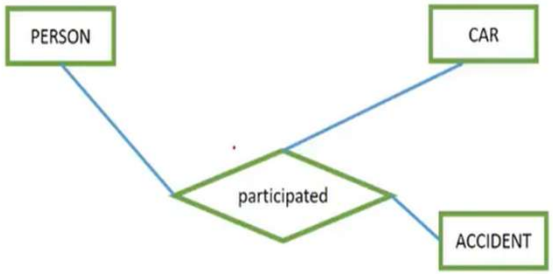
Step 4 : Identify the relationships

- Customers own one or more cars each.

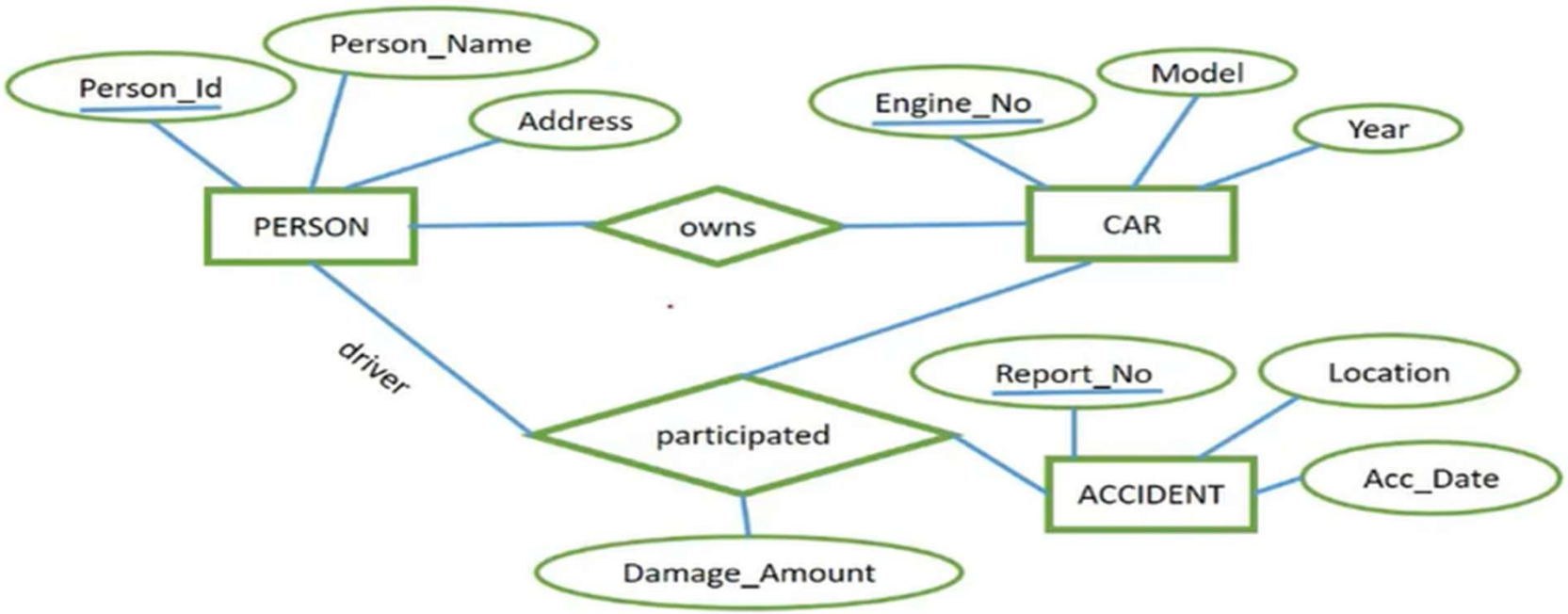


Step 4 : Identify the relationships

- Each car is associated with it zero to any number of recorded accidents.



Step 5 : Complete E R Diagram



- Design an E-R diagram for keeping track of the exploits of your favorite sports team. You should store the matches played, the scores in each match, the players in each match and individual player statistics for each match. Summary statistics should be modeled as derived attributes.

Step 1 : Identify the entity sets

- From the given question the entity sets identified are
  1. MATCH
  2. PLAYER

Step 2 : Identify the relevant attributes

- The relevant attributes of MATCH entity set are
  - Match_Id
  - Stadium
  - Date
  - Opponent
  - Own_Score
  - Opp_Score
- The relevant attributes of PLAYER entity set are
  - Player_Id
  - Player_Name
  - Summary_Score

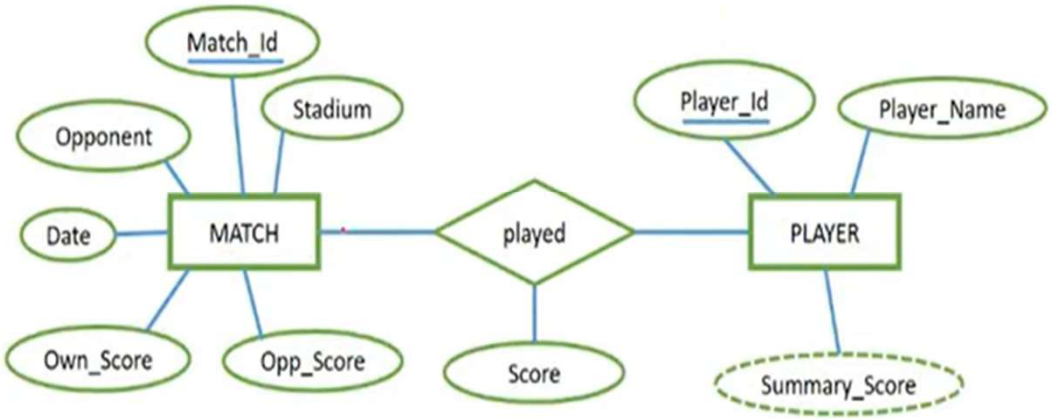
Step 3 : Identify the prime attributes

- The prime attribute of MATCH entity set is
  - Match_Id
- The prime attribute of PLAYER entity set is
  - Player_Id

Step 4 : Identify the relationships



Step 5 : Complete E R Diagram



A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; and (d) instructors, including identification number, name, department, and title. Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.

Step 1 : Identify the entity sets

- From the given question the entity sets identified are
  1. COURSE
  2. COURSE-OFFERINGS
  3. STUDENT
  4. INSTRUCTOR

Step 2 : Identify the relevant attributes

- The relevant attributes of **COURSE** entity set are
  - Course_No
  - C_Tile
  - Credits
  - Syllabus
- The relevant attributes of **COURSE-OFFERINGS** entity set are
  - Year
  - Time
  - Secno
  - Room
  - Semester
- The relevant attributes of **STUDENT** entity set are
  - S_Id
  - S_Name
  - Program
- The relevant attributes of **INSTRUCTOR** entity set are
  - I_id
  - I_Name
  - I_Title
  - Dept

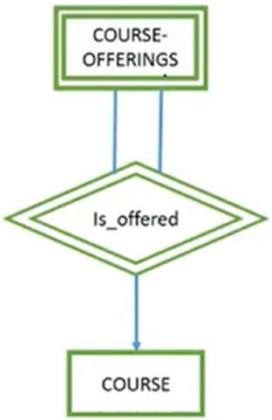
Step 3 : Identify the prime attributes

- The prime attribute of **COURSE** entity set are
  - Course_No
- There is no prime attribute in **COURSE-OFFERINGS** entity set, so it is an weak entity set, however the partial key (discriminator) is
  - Year
- The prime attribute of **STUDENT** entity set is
  - S_Id
- The prime attribute of **INSTRUCTOR** entity set is
  - I_id

Step 4 : Identify the relationship



Step 4 : Identify the relationship



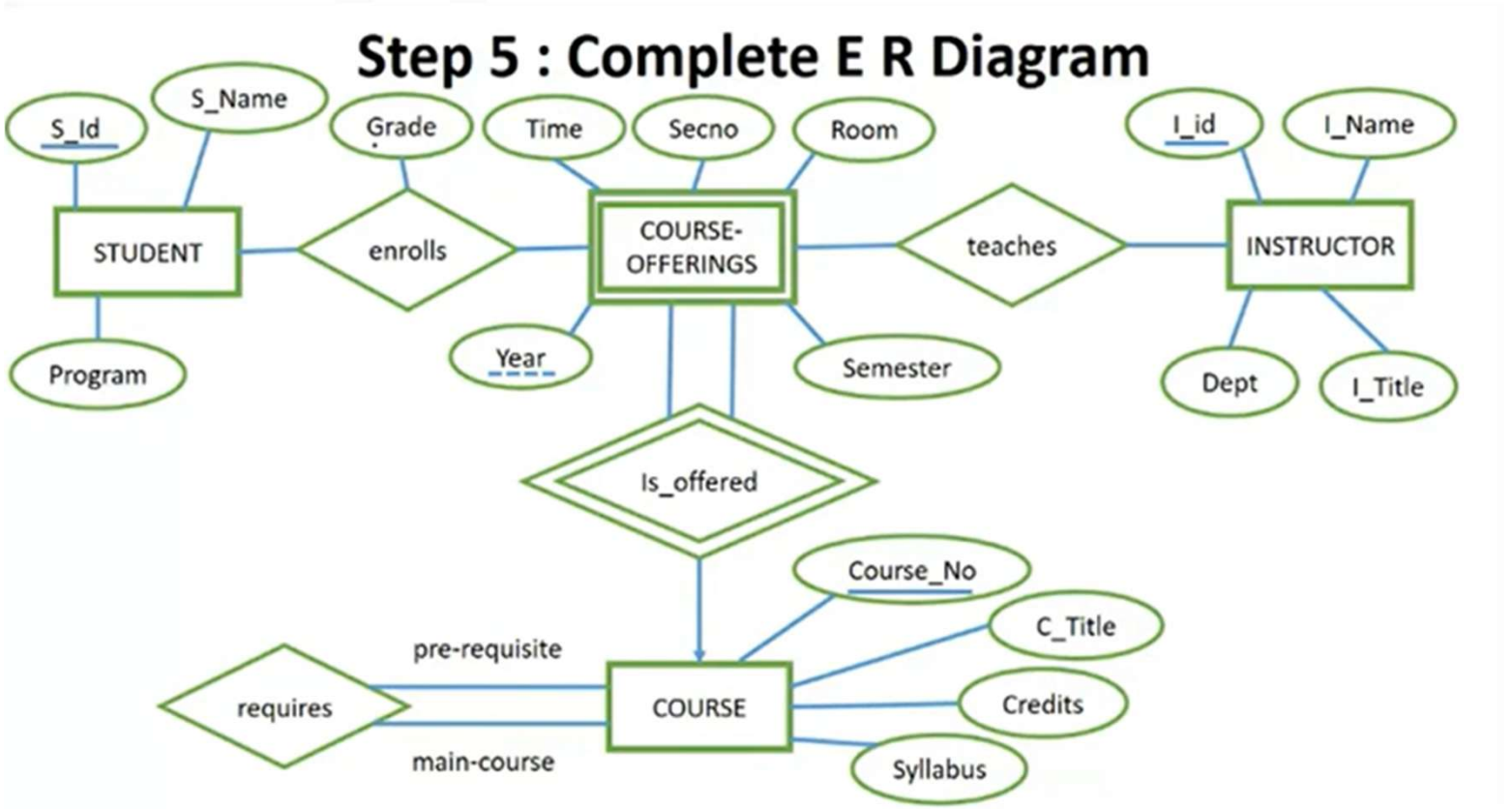
Step 4 : Identify the relationship



Step 4 : Identify the relationship



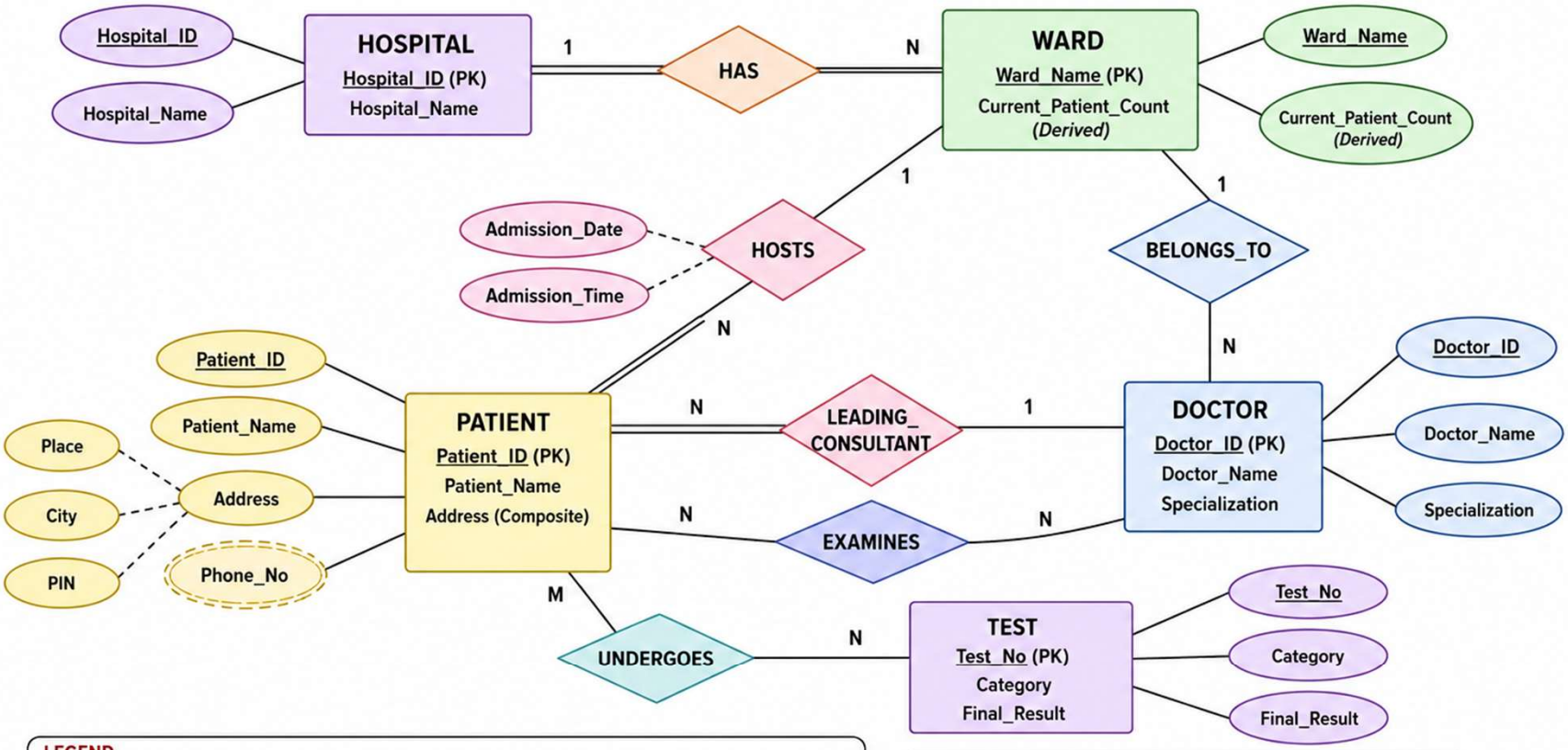
Step 5 : Complete E R Diagram



Construct a neat ER diagram for the hospital database using the following information:

A hospital has multiple specialized wards each identified by a unique name and tracking the total count of current patients. A patient has a unique ID, name, address (place, city, pin), and phone numbers. Patients undergo multiple tests, and each test has a unique test number, category, and final result. Not all patients are advised for tests. A Ward hosts multiple patients. The admission date and time are recorded upon entry. Each ward must have at least one patient. A doctor specialized to/ belongs to one specific ward but each ward can have multiple doctors. Not all wards have specialized doctors. Each patient is assigned to one specific doctor/leading consultants, but other doctors may examine them if needed. Only some doctors are acting as leading consultant for the patients.

ER DIAGRAM OF HOSPITAL DATABASE



LEGEND

Entity · Relationship · Attribute · Multivalued Attribute · Composite Attribute · Derived Attribute

PARTICIPATION CONSTRAINT

Total Participation (Mandatory)

Partial Participation (Optional)

CARDINALITY

1 : One
N : Many
M : Many

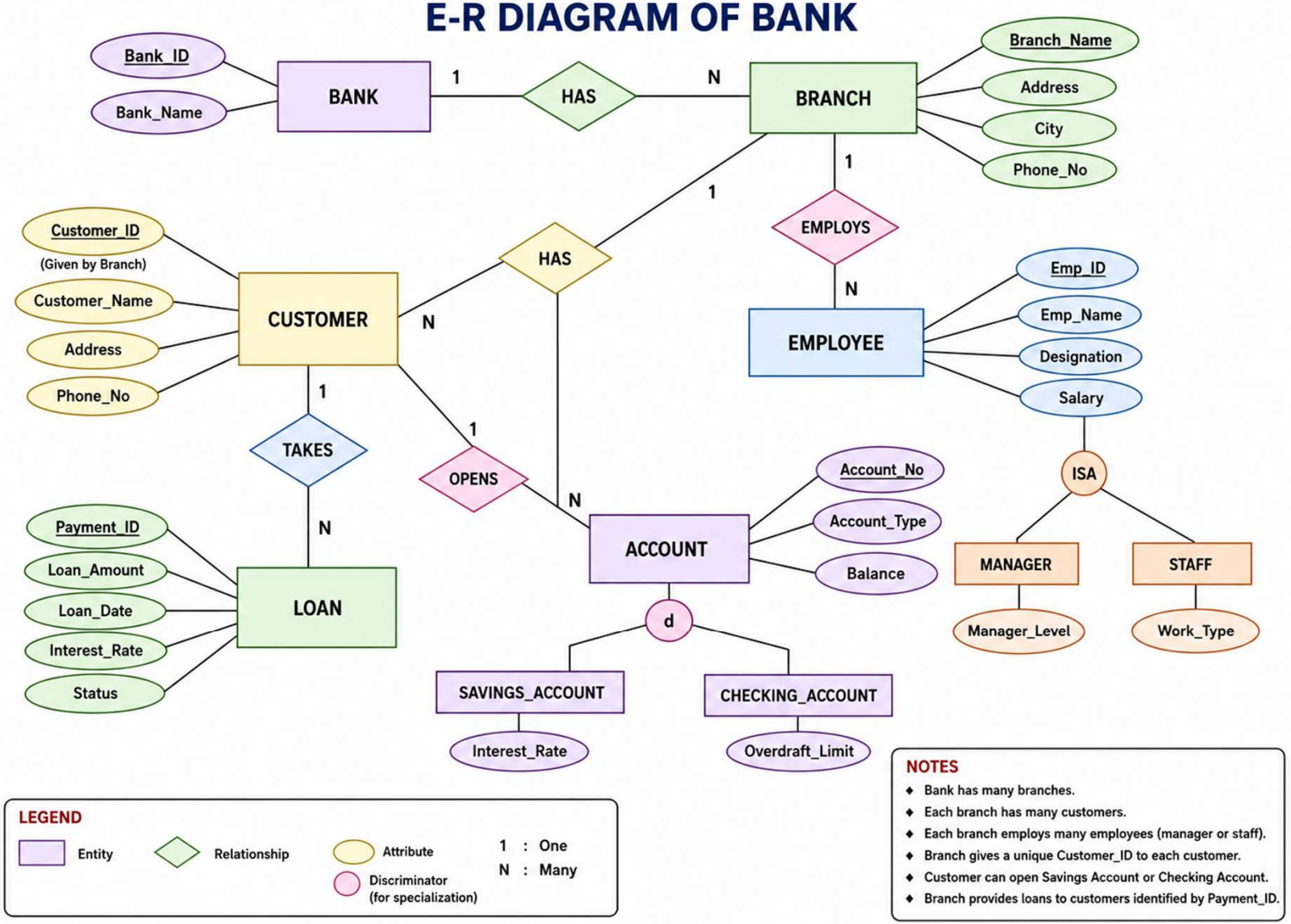
IMPORTANT CONSTRAINTS

- Each ward must have at least one patient.
- A patient is assigned to one leading consultant (Leading Consultant relationship).
- Not all patients are advised for tests (Undergoes – partial participation of Patient).
- Not all wards have specialized doctors (Belongs_To – partial participation of Ward).
- Only some doctors act as leading consultants (Leading Consultant – partial participation of Doctor).
- Other doctors may examine patients if needed (Examines – many-to-many, partial participation).

Design an E-R diagram with the help of below details:

A bank has many branches in a country.
Each branch is identified with the help of branch name.
Each branch of the bank has multiple customers.
An employee of bank can be a manager or staff of that branch.
Branch gives each Customer a unique id to identify them.
Customers can open two type of accounts i.e. Savings account and Checkingaccount. Branches of the bank provides loan to the customers which is identified with the help of payment_id.

E-R DIAGRAM OF BANK



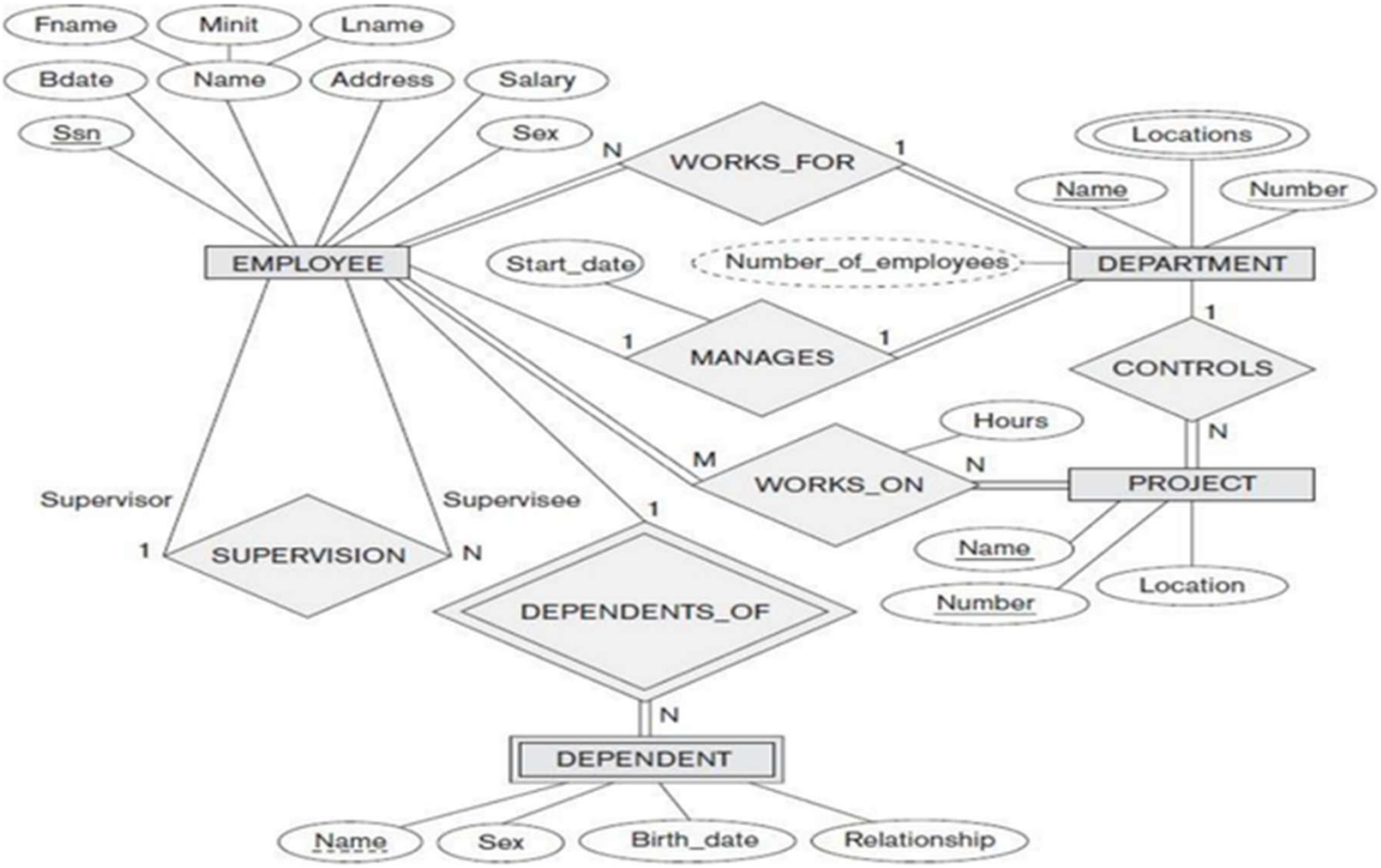
LEGEND

- Entity
- Relationship
- Attribute
- Discriminator (for specialization)
- 1 : One
- N : Many

NOTES

- Bank has many branches.
- Each branch has many customers.
- Each branch employs many employees (manager or staff).
- Branch gives a unique Customer_ID to each customer.
- Customer can open Savings Account or Checking Account.
- Branch provides loans to customers identified by Payment_ID.

A Company contains several Departments. Each Department have several Locations. Departments are identified by a department name, Department no and Department Locations. Managers control a particular department. Each Department is associated with number of projects. Each of which has a uniquename, a unique number, and a single location. Employees are identified by name, id, address, dob, phone no, date_of_joining and years of experience. We will keep track of the direct supervisor of each employee (who is another employee). An employee belongs to only one department but can work on several project. We also keep track of number of hours worked by an employee on a single project. Each employee has dependent. Dependents have D_name, Gender, and relationship.


Fname
Minit
Lname
Bdate
Name
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
Start_date
Number_of_employees
DEPARTMENT
1
1
MANAGES
1
CONTROLS
Hours
N
M
WORKS_ON
N
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship

Hospital Management System

A hospital maintains information about **Doctors, Patients, Departments, Appointments, and Treatments**.

•Each **Department** has a unique Department_ID, Department_Name, and Location.

•Each **Doctor** has Doctor_ID, Name, Qualification, Specialization, Phone_No, and Salary. Every doctor works in exactly one department.

•A department may have several doctors.

•Each **Patient** has Patient_ID, Name, Address, Age, Gender, Blood_Group, and Phone_No.

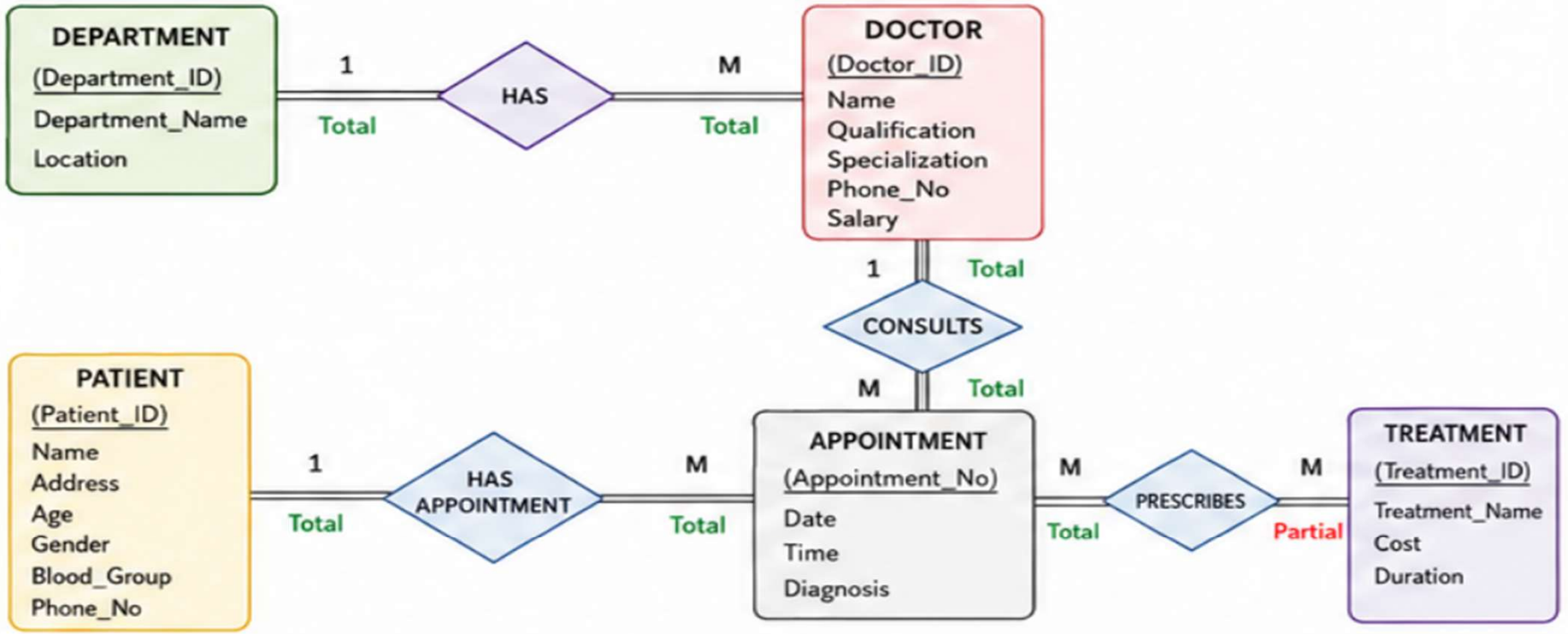
•A patient can consult many doctors, and a doctor can treat many patients.

•Every consultation is recorded as an **Appointment**, having Appointment_No, Date, Time, and Diagnosis.

•During each appointment, one or more **Treatments** may be prescribed. Each treatment has Treatment_ID, Treatment_Name, Cost, and Duration.

•Each patient may have multiple appointments, while each appointment belongs to only one patient.

Q1. HOSPITAL MANAGEMENT SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

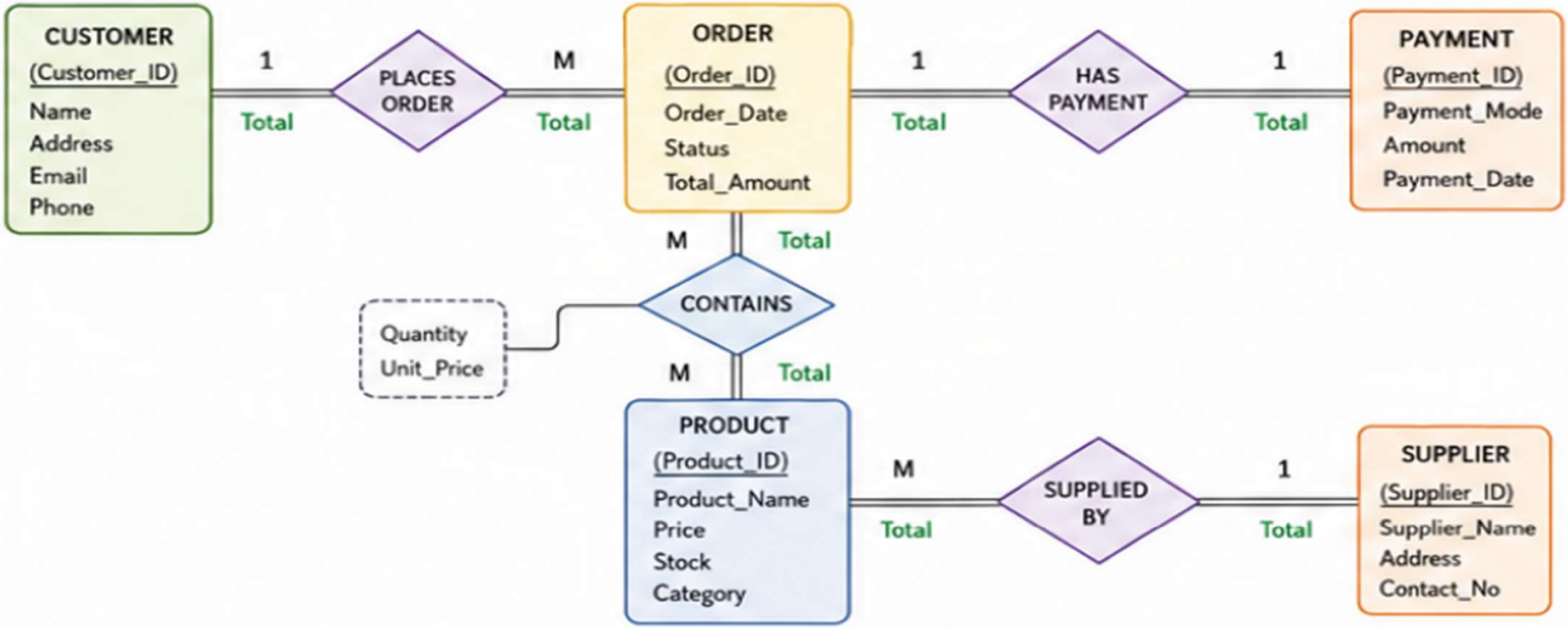
• DEPARTMENT HAS DOCTOR	1 : M	(Total — Total)
• DOCTOR CONSULTS APPOINTMENT	1 : M	(Total — Total)
• PATIENT HAS APPOINTMENT	1 : M	(Total — Total)
• APPOINTMENT PRESCRIBES TREATMENT	M : M	(Total — Partial)

Online Shopping System

An e-commerce company maintains information about **Customers, Products, Orders, Suppliers, and Payments**.

- Each **Customer** has Customer_ID, Name, Address, Email, and Phone.
- Each customer can place many orders, but every order belongs to exactly one customer.
- Each **Order** has Order_ID, Order_Date, Status, and Total_Amount.
- Every order contains one or more products.
- Each **Product** has Product_ID, Product_Name, Price, Stock, and Category.
- A product can appear in many orders.
- The quantity of each product ordered should also be recorded.
- Products are supplied by suppliers.
- Each **Supplier** has Supplier_ID, Supplier_Name, Address, and Contact_No.
- One supplier supplies many products, but every product is supplied by only one supplier.
- Every order has one payment having Payment_ID, Payment_Mode, Amount, and Payment_Date.

Q2. ONLINE SHOPPING SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

• CUSTOMER PLACES ORDER	1 : M	(Total — Total)
• ORDER HAS PAYMENT	1 : 1	(Total — Total)
• ORDER CONTAINS PRODUCT	M : M	(Total — Total)
• PRODUCT SUPPLIED BY SUPPLIER	M : 1	(Total — Total)

Library Management System

A university library maintains information about **Books, Authors, Members, Librarians, and Book Issues**.

•Each **Book** has Book_ID, ISBN, Title, Publisher, and Edition.

•A book may have one or more authors.

•Each **Author** has Author_ID, Name, Nationality, and Email.

•The library has many **Members**, each identified by Member_ID, Name, Department, Phone_No, and Membership_Type.

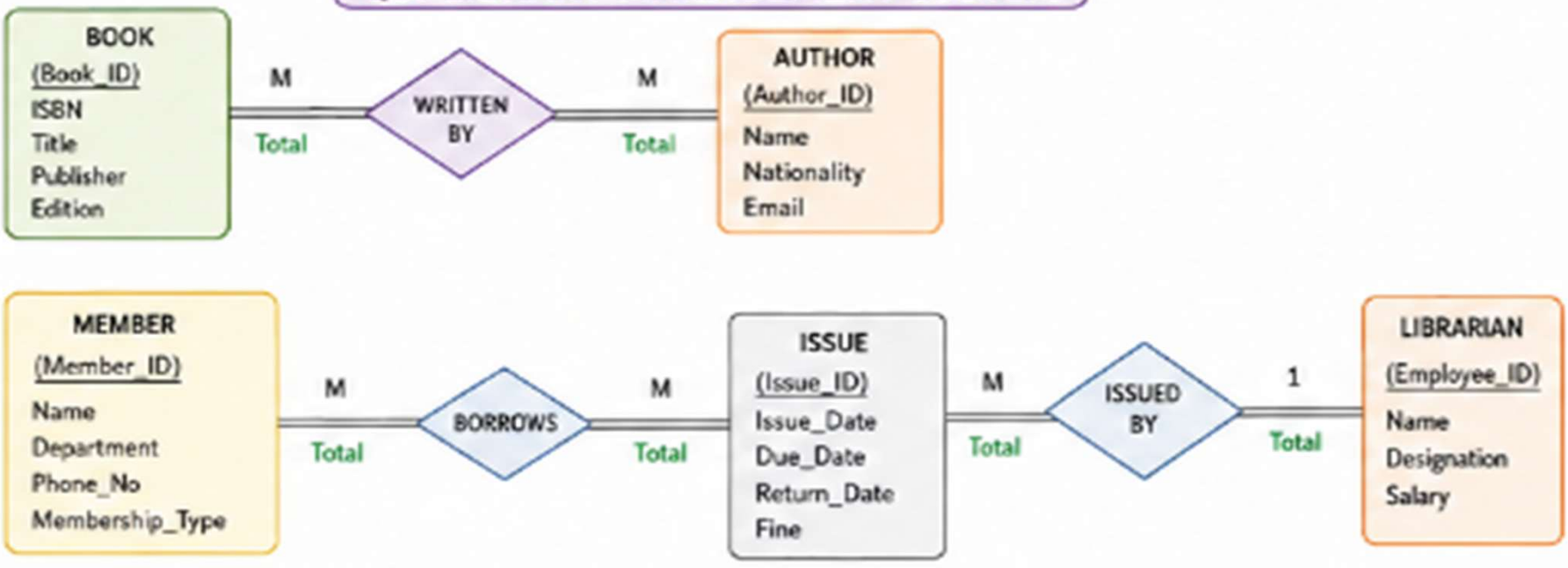
•Members can borrow many books, and each book can be borrowed by many members over time.

•The Issue relationship stores Issue_Date, Due_Date, Return_Date, and Fine.

•The library employs **Librarians**, each having Employee_ID, Name, Designation, and Salary.

•One librarian issues many books, while every issue transaction is handled by exactly one librarian.

Q3. LIBRARY MANAGEMENT SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

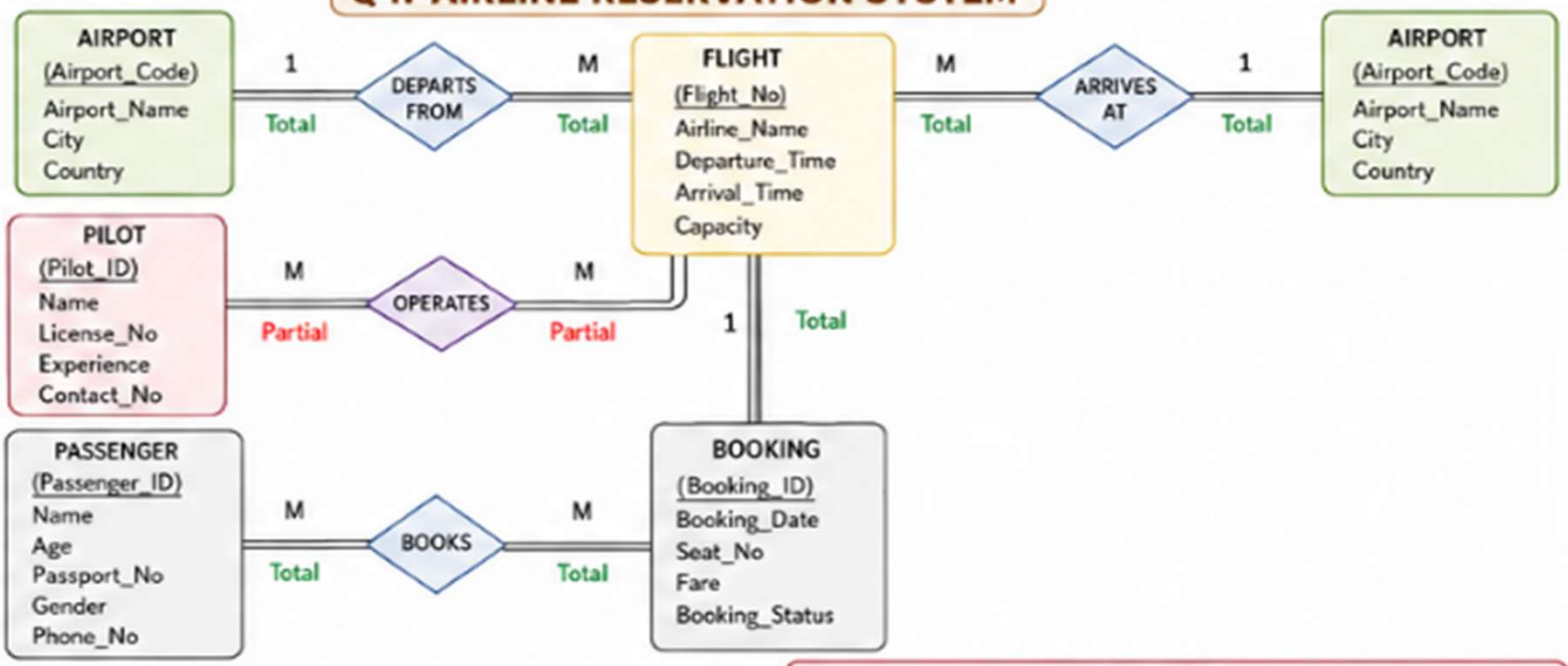
• BOOK WRITTEN BY AUTHOR	M : M	(Total — Total)
• MEMBER BORROWS BOOK (ISSUE)	M : M	(Total — Total)
• ISSUE ISSUED BY LIBRARIAN	M : 1	(Total — Total)

Airline Reservation System

An airline company maintains data about **Flights, Passengers, Pilots, Airports, and Bookings**.

- Each **Flight** has Flight_No, Airline_Name, Source, Destination, Departure_Time, Arrival_Time, and Capacity.
- Flights depart from one airport and arrive at another airport.
- Each **Airport** has Airport_Code, Airport_Name, City, and Country.
- A pilot can fly many flights, and each flight may be operated by one or more pilots.
- Each **Pilot** has Pilot_ID, Name, License_No, Experience, and Contact_No.
- Each **Passenger** has Passenger_ID, Name, Age, Passport_No, Gender, and Phone_No.
- Passengers can book many flights, and every flight can have many passengers.
- Booking information includes Booking_ID, Booking_Date, Seat_No, Fare, and Booking_Status.
- Every booking belongs to exactly one passenger and one flight.

Q4. AIRLINE RESERVATION SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

• AIRPORT DEPARTS FROM FLIGHT	1 : M	(Total — Total)
• FLIGHT ARRIVES AT AIRPORT	M : 1	(Total — Total)
• PILOT OPERATES FLIGHT	M : M	(Partial — Partial)
• PASSENGER BOOKS FLIGHT (BOOKING)	M : M	(Total — Total)
• FLIGHT HAS BOOKING	1 : M	(Total — Total)

Banking System

A bank maintains information about its **Branches, Customers, Accounts, Transactions, Loans, and Employees**.

Each **Branch** has Branch_ID, Branch_Name, Address, and City. A branch maintains many accounts, while each account belongs to only one branch.

Each **Customer** has Customer_ID, Name, Address, Phone, and Email. A customer can own multiple accounts, and an account may be jointly owned by multiple customers.

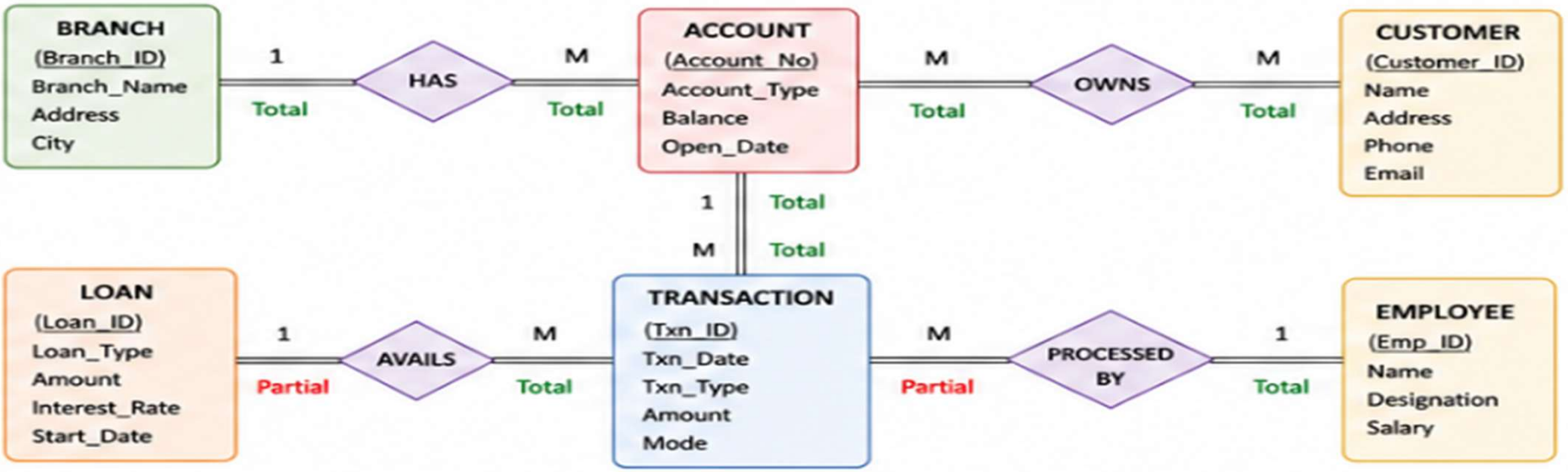
Each **Account** has Account_No, Account_Type, Balance, and Open_Date. Each account records several transactions.

Each **Transaction** has Transaction_ID, Transaction_Date, Transaction_Type, Amount, and Mode. Every transaction is processed by exactly one employee.

Each **Employee** has Employee_ID, Name, Designation, and Salary. An employee may process many transactions.

The bank also provides **Loans**. Each loan has Loan_ID, Loan_Type, Amount, Interest_Rate, and Start_Date. A customer may avail several loans, whereas each loan belongs to only one customer.

2. BANKING SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

• BRANCH HAS ACCOUNT	1 : M	(Total – Total)
• ACCOUNT OWNS CUSTOMER	M : M	(Total – Total)
• ACCOUNT HAS TRANSACTION	1 : M	(Total – Total)
• ACCOUNT AVAILS LOAN	M : 1	(Total – Partial)
• TRANSACTION PROCESSED BY EMPLOYEE	M : 1	(Partial – Total)

Hotel Management System

A hotel maintains information about its **Hotels, Rooms, Guests, Employees, and Services**.

Each **Hotel** has Hotel_ID, Hotel_Name, Address, City, and Phone. Every hotel contains several rooms and employs many employees.

Each **Room** has Room_No, Room_Type, Tariff, and Status. Every room belongs to exactly one hotel.

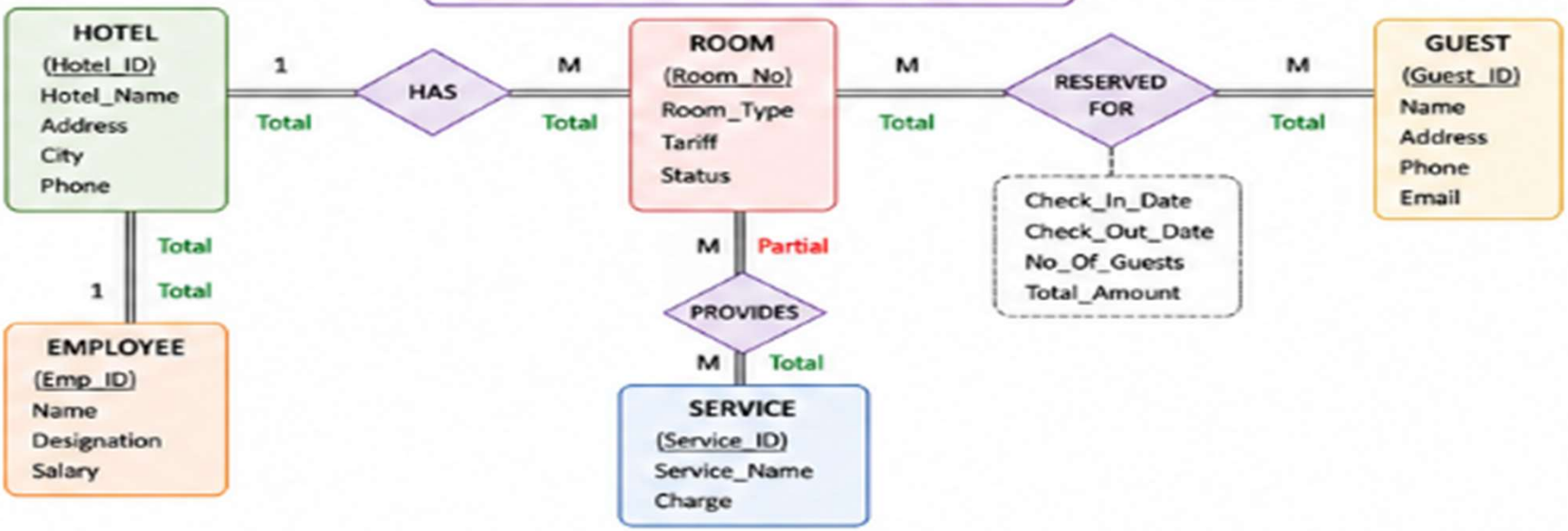
Each **Guest** has Guest_ID, Name, Address, Phone, and Email. Guests reserve rooms. A guest may reserve several rooms over time, and a room may be reserved by many guests at different times.

For every reservation, the hotel records **Check_In_Date, Check_Out_Date, Number_of_Guests, and Total_Amount**.

Each **Employee** has Employee_ID, Name, Designation, and Salary. Employees work in only one hotel.

The hotel also provides several **Services** such as Laundry, Wi-Fi, Restaurant, and Spa. Each service has Service_ID, Service_Name, and Charge. A room may receive many services, and one service can be provided to many rooms.

3. HOTEL MANAGEMENT SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

• HOTEL HAS ROOM	1 : M	(Total – Total)
• ROOM RESERVED FOR GUEST	M : M	(Total – Total)
• HOTEL EMPLOYS EMPLOYEE	1 : M	(Total – Total)
• ROOM PROVIDES SERVICE	M : M	(Partial – Total)

Transport Management System

A transport company maintains information about **Transport Companies, Vehicles, Drivers, Trips, Passengers, and Bookings**.

Each **Transport Company** has Company_ID, Company_Name, Address, and Contact_No. A company owns several vehicles, while every vehicle belongs to only one company.

Each **Vehicle** has Vehicle_No, Type, Capacity, Model, and Status. Vehicles are assigned to drivers. A driver may drive different vehicles over time, while a vehicle is assigned to one driver at a particular time.

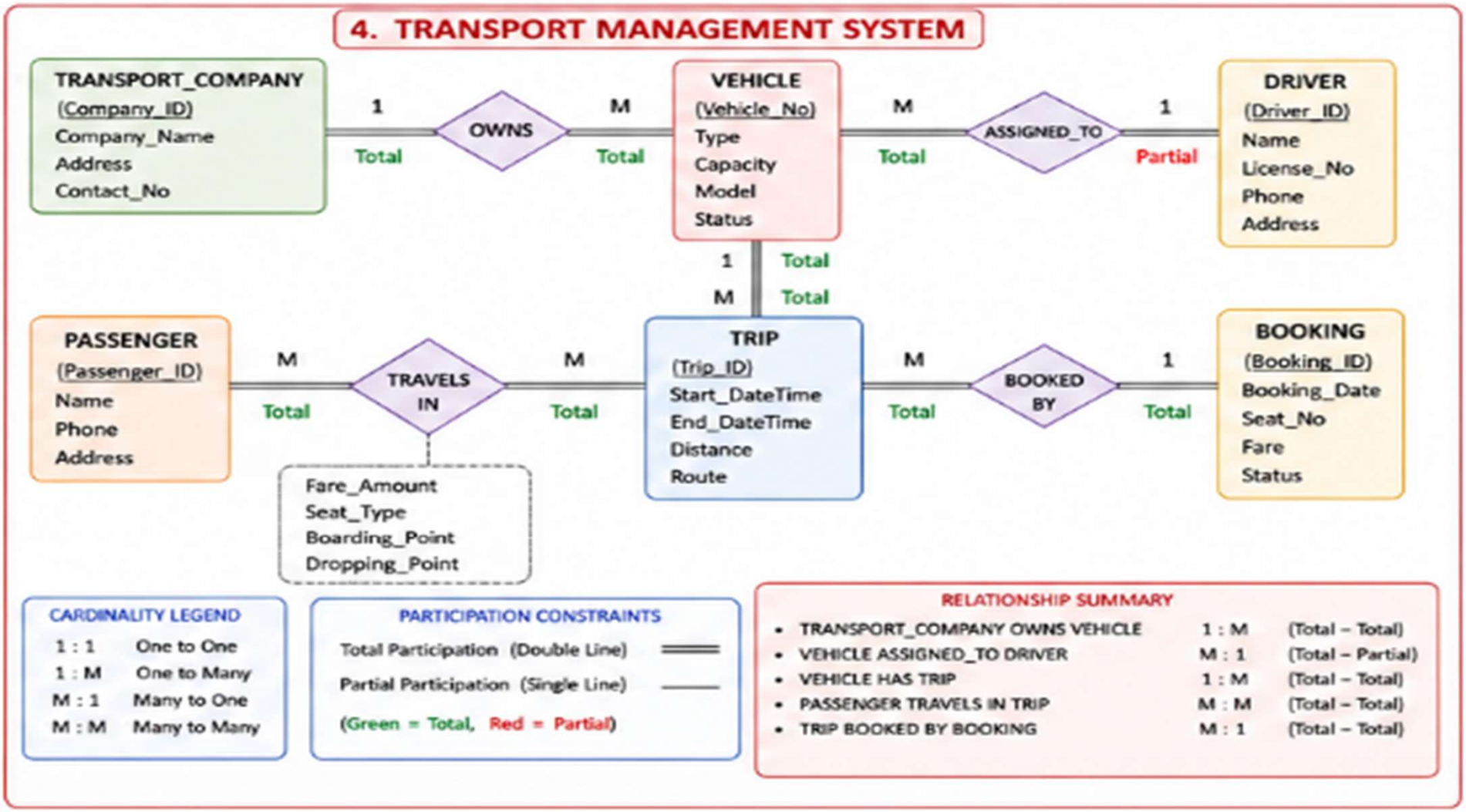
Each **Driver** has Driver_ID, Name, License_No, Phone, and Address.

Each **Trip** has Trip_ID, Start_Date_Time, End_Date_Time, Distance, and Route. A vehicle performs several trips.

Each **Passenger** has Passenger_ID, Name, Phone, and Address. Passengers travel on trips. A passenger can travel on many trips, and each trip may carry many passengers.

The booking information includes **Booking_ID, Booking_Date, Seat_No, Fare, and Status**. Each booking is associated with exactly one trip.

4. TRANSPORT MANAGEMENT SYSTEM



CARDINALITY LEGEND

1 : 1	One to One
1 : M	One to Many
M : 1	Many to One
M : M	Many to Many

PARTICIPATION CONSTRAINTS

Total Participation (Double Line) ═══

Partial Participation (Single Line) ———

(Green = Total, Red = Partial)

RELATIONSHIP SUMMARY

• TRANSPORT_COMPANY OWNS VEHICLE	1 : M	(Total – Total)
• VEHICLE ASSIGNED_TO DRIVER	M : 1	(Total – Partial)
• VEHICLE HAS TRIP	1 : M	(Total – Total)
• PASSENGER TRAVELS IN TRIP	M : M	(Total – Total)
• TRIP BOOKED BY BOOKING	M : 1	(Total – Total)